3.2 Classwork: Transformations of Trigonometric Functions — Markscheme

1. [Maximum mark: 10]

   $f(x) = 4\sin\left(2\left(x - \frac{\pi}{6}\right)\right) - 1$

   (a) Vertical stretch, scale factor 4 **A1**

   Horizontal stretch, scale factor $\frac{1}{2}$ **A1**

   Horizontal translation $\frac{\pi}{6}$ to the right **A1**

   Vertical translation 1 down **A1**

   "Describe fully" needs words (markscheme-spec §19). One mark per transformation, each with its type, direction and size. Accept "horizontal compression by a factor of 2" for the horizontal stretch. The vertical translation must say "down". Accept "translation by $\binom{\pi/6}{-1}$" for both translation marks. Reject "move". A correct chain of equations with no words earns at most 1 mark.

   Order: the horizontal stretch must come before the translation of $\frac{\pi}{6}$ right, and the vertical stretch before the translation down. Also fully correct: translate $\frac{\pi}{3}$ right, then stretch horizontally with factor $\frac{1}{2}$. Deduct 1 mark for a wrong order. Wrong methods, each A0 for that transformation: translation $\frac{\pi}{6}$ left; horizontal stretch with factor 2.

   (b) amplitude $= 4$ **A1**

   period $= \dfrac{2\pi}{2} = \pi$ **A1**

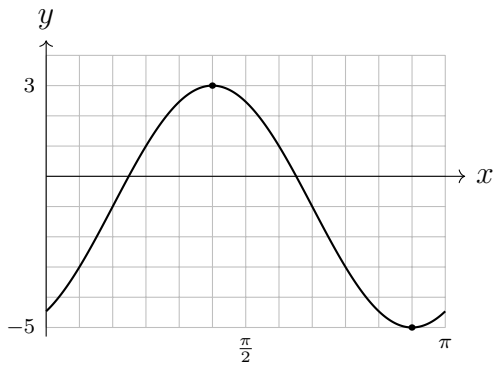
   range: $-5 \le y \le 3$ **A1**

   Accept $[-5,\ 3]$ or $-5 \le f(x) \le 3$. Strict inequalities earn A0. Wrong methods: period 4π ($2\pi \times 2$) earns A0; range $-4 \le y \le 4$ (vertical translation ignored) earns A0.

   (c) Sine-wave shape on $0 \le x \le \pi$, starting and ending at $y \approx -4.46$, one maximum and one minimum **A1**

   Maximum $\left(\dfrac{5\pi}{12},\ 3\right)$ **A1**

   Minimum $\left(\dfrac{11\pi}{12},\ -5\right)$ **A1**

   First A1: smooth curve, correct shape, endpoints near -4.5, with the maximum and minimum within half a grid square of the correct positions. Each coordinate pair must be exact; decimals from the GDC ($(1.31,\ 3)$, $(2.88,\ -5)$) earn A0 for that point. Wrong methods: translating left gives $\left(\frac{\pi}{12},\ 3\right)$ and $\left(\frac{7\pi}{12},\ -5\right)$; solving $2x - \frac{\pi}{6} = \frac{\pi}{2}$ (bracket dropped) gives $x = \frac{\pi}{3}$. Each earns A0 for that point. FT the coordinates from the candidate's range in (b).

2. [Maximum mark: 11]

$D(t) = a\cos\big(b(t-c)\big) + d$, $0 \le t \le 12$. Maximum 15.1 at $t = 5.7$; minimum 9.3 at $t = 11.7$.

(a) $a = \dfrac{15.1 - 9.3}{2}$ or $d = \dfrac{15.1 + 9.3}{2}$ **M1**

$a = 2.9$ **A1**

$d = 12.2$ **A1**

M1 for either formula. Correct values with no working earn M0A1A1.

(b) period $= 2(11.7 - 5.7) = 12$, so $b = \dfrac{2\pi}{12}$ **M1**

$b = \dfrac{\pi}{6}$ $\quad (0.523598\ldots)$ **A1**

$c = 5.7$ **A1**

Accept $b = 0.524$ or the display with an ellipsis. For c: the cosine has its maximum where $t - c = 0$, so c is the time of the longest day. Wrong methods, each A0: $b = 12$ (period used as b); $c = 11.7$ (time of the minimum); $c = 2.7$ (quarter-period shift, as for sine); $c = -5.7$ (sign).

(c) $D(10.1) = 2.9\cos\left(\dfrac{\pi}{6}(10.1 - 5.7)\right) + 12.2$ **M1**

$= 10.3$ hours $\quad (10.2595\ldots)$ **A1**

A1 for 10.3 or the display with an ellipsis; 10.26 with no ellipsis earns A0 (markscheme-spec §21). Wrong method: degree mode gives 15.1 (15.0976…), earning M1A0. FT from the candidate's a, b, c, d.

(d) $2.9\cos\left(\dfrac{\pi}{6}(t - 5.7)\right) + 12.2 = 14$ **M1**

GDC: graph $y = D(t)$ and $y = 14$; find both intersections.

$t = 3.98$ $(3.97888\ldots)$ and $t = 7.42$ $(7.42111\ldots)$ **A1**

length $= 7.42111\ldots - 3.97888\ldots = 3.44$ months $\quad (3.44223\ldots)$ **A1**

M1 for the equation or inequality $D(t) > 14$, or a sketch with the line $D = 14$ drawn. The first A1 needs both values. Accept 3.44 from the rounded values, $7.42 - 3.98$. Wrong methods: giving one intersection as the answer (3.98 or 7.42) earns M1A0A0; degree mode gives no solutions in $0 \le t \le 12$ and earns M1A0A0. FT from the candidate's model.